

# PH 716 Applied Survival Analysis

## Part 10: Recurent Events

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### Motivation

- Standard survival analysis
  - Focus: time to first event
  - Example: time to death, first relapse
- Recurrent event setting
  - Individuals can experience multiple events over time
  - Examples: hospital readmissions, asthma attacks, tumor recurrences, equipment breakdowns

### A concrete example: `survival::bladder1`

Full dataset from a study on recurrences of bladder cancer. It contains all three treatment arms and all recurrences for 118 subjects; the maximum number of recurrences is 9.

- `id`: Patient id
- `treatment`: Placebo, pyridoxine (vitamin B6), or thiotepa
- `number`: Initial number of tumours
- `size`: size (cm) of largest initial tumour
- `recur`: Number of recurrences
- `start`, `stop`: The start and end time of each time interval
- `status`: End of interval code (0=censored, 1=recurrence, 2=death from bladder disease, 3=death other/unknown cause)
- `rtumor`: Number of tumors found at the time of a recurrence
- `rsize`: Size of largest tumor at a recurrence
- `enum`: Event number within patient

```
head(survival::bladder1)
```

```
##   id treatment number size recur start stop status rtumor rsize enum
## 1  1  placebo      1    1     0     0   0      3      .      .    1
## 2  2  placebo      1    3     0     0   1      3      .      .    1
## 3  3  placebo      2    1     0     0   4      0      .      .    1
## 4  4  placebo      1    1     0     0   7      0      .      .    1
## 5  5  placebo      5    1     0     0  10      3      .      .    1
## 6  6  placebo      4    1     1     0   6      1      1      1    1
```

### Why standard Cox model is insufficient

The usual Cox model assumes:

- Only one event per subject
- Independence across observations

But for recurrent events:

- Multiple events per subject
- Within-subject dependence
- Risk may change after each event

## Notations for recurrent events

- $T_{ik}$ : authentic time of the  $k$ th event for subject  $i$
- $N_i(t)$ : number of events that have happened up to time  $t$ , forming a **counting process**
- $Y_i(t)$ : at-risk indicator, indicating whether subject  $i$  is under observation just before time  $t$
- $x_{ij}$ : the  $j$ th covariate for subject  $i$

## Counting process

Imagine each person carries a counter:

- Every time an event happens  $\rightarrow$  the counter goes up by 1
- Between events  $\rightarrow$  the counter stays the same

That counter is a counting process:

- In time-to-first-event: the counter can jump at most once.
- In recurrent events: it looks like a step function: flat  $\rightarrow$  jump  $\rightarrow$  flat  $\rightarrow$  jump  $\rightarrow$  flat  $\rightarrow$  jump

We track:

- In time-to-first-event: “How fast does the first jump happen?” (i.e., the hazard function)
- In recurrent events: “How fast do the jumps happen over time?” (i.e., the **intensity function**, an extension of the hazard function)

## Intensity function

- Define the increment over a tiny time interval  $[t, t + dt)$ ,  $dt > 0$

$$dN_i(t) = N_i(t + dt) - N_i(t)$$

Then  $dN_i(t)$  is an indicator:

- $dN_i(t) = 1$  if an event happens in  $[t, t + dt)$
- $dN_i(t) = 0$  if no event happens in  $[t, t + dt)$
- Reducing to the usual failure indicator in time-to-first-event setting
- The counting process  $N_i(t)$  is characterized via its intensity:
 
$$\lambda_i(t) = \lim_{dt \rightarrow 0} \frac{\Pr(dN_i(t) = 1 \mid \mathcal{F}_{t-})}{dt}$$
  - $\mathcal{F}_{t-}$ : the history of events, censoring, and covariates right before time  $t$
  - Interpretation: the instantaneous event rate at time  $t$ , given the past history of events and covariates
  - Extension of the hazard function to recurrent events
- Expected cumulative number of events by time  $t$

$$E\{N_i(t)\} = \int_0^t \lambda_i(u) du$$

- Probability of no recurrent event occurring by time  $t$

$$\Pr(N_i(t) = 0) = \exp\left(-\int_0^t \lambda_i(u) du\right)$$

## Andersen-Gill model

- Assume:
  - Recurrent events as repeated jumps of the same counting process
  - No within-subject dependence
- Andersen-Gill model

$$\lambda_i(t) = Y_i(t)\lambda_0(t) \exp \left( \sum_{j=1}^p x_{ij}\beta_j \right)$$

- $\lambda_0(t)$ : baseline intensity function
- $\beta_j$ : regression coefficient for covariate  $j$

### Ex. 10.1 Revisit `survival::bladder1`

Only recurrences (`status==1`) count as events in the counting process. Death stops observation, acting like terminal censoring.

```
options(digits=4)
library(survival)
df = subset(survival::bladder1, stop > start)
df$event = ifelse(df$status==1, 1, 0)
df$Treatment = factor(df$treatment)
df$number = factor(df$number)
df$rsiz = factor(df$rsiz)
# Andersen-Gill model
## Accounting for within-subject dependence
fit_ag_1 = coxph(
  Surv(start, stop, event) ~ Treatment + number + rsiz,
  data = df
)
summary(fit_ag_1)
## Accounting for within-subject dependence (via robust variance estimation)
fit_ag_2 = coxph(
  Surv(start, stop, event) ~ Treatment + number + rsiz + cluster(id),
  data = df
)
summary(fit_ag_2)

# Expected cumulative number of events up to time 20 for a new subject
base = basehaz(fit_ag_2, centered = FALSE) # baseline cumulative intensity function
newx = data.frame(
  Treatment = factor("thiotepa", levels = levels(df$Treatment)),
  number = factor(2, levels = levels(df$number)),
  rsiz = factor(2, levels = levels(df$rsiz))
)
lp = predict(fit_ag_2, newdata = newx, type = "lp")
(expected_num_events = exp(lp) * max(base$hazard[base$time <= 20]))

# Further give the probability of no recurrent event occurring by time 20 for the same subject
(prob_no_event = exp(-expected_num_events))
```